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Paleomagnetism of Lower Cretaceous Lavas on Jamaica

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With 2 Figures

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Summary

A small Lower Cretaceous igneous province on Jamaica has been assigned an Albian or Aptian age on the basis of the fauna of associated sediments. Despite pronounced tropical weathering it was possible to find three reasonably fresh lava exposures, paleomagnetic investigations of which suggest that during the Lower Cretaceous period the southern polarity geomagnetic pole, as viewed from Jamaica, was situated in the present day northeast Pacific. An interpretation of this result in terms of plate tectonics yields a movement of the Caribbean tectonic plate over a distance of 1000 km to the northeast since Lower Cretaceous. This result is in good agreement with similar deductions made from previously investigated Upper Cretaceous rocks on Jamaica. It suggests also that no appreciable movement of the Caribbean plate or polar wonder had occurred during the Cretaceous age. One of the three lava units reveals an equatorial direction of magnetization also found previously in Upper Cretaceous rocks of the island. This direction could have been caused by a non axial dipole configuration during a reversal of the geomagnetic field. This interpretation is supported by other evidence of a field reversal at the close of Aptian and beginning of Albian.

Zusammenfassung

Paläomagnetische Untersuchung an Lavaströmen der Unterkreide auf Jamaika

Auf Jamaika wurden Vulkanite aus der Unterkreide paläomagnetisch untersucht, die gemäß der Fauna der zugehörigen Sedimente in die albische oder aptische Zeit eingeordnet werden können. Trotz großer Schwierigkeiten mit tro-

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pischer Verwitterung, Geländeform und Vegetation konnten Bohrkerne von drei verschiedenen Lavaeinheiten gewonnen werden. Unter Anwendung der üblichen Labormethoden wurde daraus die Paläomagnetfeldrichtung in der Unterkreide bestimmt.

Das Ergebnis kann nach der Theorie der Plattentektonik so gedeutet werden, daß sich Jamaika mit der gesamten karibischen Platte seit der Kreidezeit um 1000 km nach Nordosten bewegt hat. Ähnliche Resultate wurden bereits früher für die Oberkreide ebenfalls durch eine Untersuchung auf Jamaika gewonnen, welche somit durch die hier vorgelegten Datenangaben bestätigt wird.

Eine Lavaeinheit ergab eine außergewöhnliche Pollage in äquatorialen Breiten. Derartige Pollagen wurden ebenfalls früher auf Jamaika bei vulkanischen Gesteinen der Oberkreide beobachtet und sind auch bei anderen paläomagnetischen Untersuchungen manchmal festgestellt worden. Das Phänomen wird als „Momentaufnahme" einer geomagnetischen Feldumkehr gedeutet, bei der die Umpolung offensichtlich unter Beibehaltung des Dipolcharakters als eine Art „Umklapp"-Vorgang erfolgt. Für diese Interpretation spricht, daß gerade am Übergang vom Alb zum Apt eine der seltenen Feldumkehrungen der Kreidezeit beobachtet wurde.

1. Introduction

Conflicting hypotheses exist concerning the origin of the Carribean arc and its geologic history. The spectrum of proposed hypotheses includes such concepts as large scale rotations of the area or its component parts [1, 2], translational motions in various directions and over differing distances [3, 4] as well as views opposed to the hypothesis of mobile plate behavior of the Carribean [5].

Paleomagnetic data could be used to test these hypotheses, but so far only little paleomagnetic work has been done in the Caribbean. Recently, however, paleomagnetic investigations of Jamaica have been performed and this report deals with that part of the results which are concerned with Lower Cretaceous lava flows on the island.

2. Geology

Rocks of Cretaceous age probably underlie the whole of Jamaica. They are overlain by a covering of Tertiary rocks through which they appear in the form of various inliers, usually built up of rocks of Upper Cretaceous age. Within one of these inliers rocks of Lower Cretaceous age are also outcropping. This socalled Benbow Inlier is situated in the north-northeastern central region of the island. The inlier consists of limestones containing Albian and Aptian fauna, interbedded with conglomerates, tuffs, and lava flows. The area has been folded in post-Cretaceous time on an east-west axis and

the major anticline has dips of maximum 40°. The fault systems within the inlier are mainly oriented in the east-west and northwest-southeast directions. The lavas are exposed on the northern flank of the anticline and the dips of these Cretaceous beds range from 15° to 30° to the north [6].

3. Sampling

The terrain of the Benbow Inlier is mountainous and rather inaccessible. Because of the marine tropical climate and thick vegetation the choice of suitable sampling sites was extremely difficult. A thick weathered talus occurs in the region of interest and the rarely exposed rocks have been subject to pronounced tropical weathering. Nevertheless three acceptable sites could be located and were sampled successfully. The data relating to the sampling sites are given in Tab. 1.

Table 1. *Description of Sampling Sites*

Site	Location	No. of Samples	Dip	Lava Material	Weathering
W	18.3°N, 77.0°W	14	25° to N	diabase	fresh
X	18.2°N, 77.0°W	15	15° to N	diabase	rather weathered
Y	18.2°N, 76.9°W	13	25° to N	andesite	rather weathered

The samples were obtained using a portable gasoline powered diamond drill and were oriented *in situ* with a magnetic compass. Since the declination of the earth's magnetic field in Jamaica is negligible there was no need to apply a correction to the azimuths measured relative to the local magnetic meridian. Tectonic corrections were applied by measuring the dips of the sedimentary layers in the vicinity of the sampling sites.

4. Experimental Methods

The direction and magnitude of the intensity of magnetization in the specimens were measurend on a Princeton Applied Research (PAR) spinner magnetometer, model SM-1. The error of measurement was less than 10^{-7} emu/cm^3 in magnitude of intensity of magnetization and accordingly 1° in direction. Because of the existence of secondary components of the natural remanent magnetization (NRM) the specimens were subjected to magnetic and thermal "cleaning" procedures. Alternating field (AF) demagnetization was conducted using

an equipment with a three axis tumbler and capable of generating peak fields up to 5000 oe. Thermal demagnetization was performed using the apparatus described by Vincenz et al. [7].

5. Magnetic Minerals

The rocks exposed at sites *W* and *X* (Tab. 1) consist of diabases and polished section microscopy showed that the NRM was associated with large grains of homogeneous titanomagnetite. The grain diameters ranged from 200—500 μ. The homogeneous titanomagnetite revealed a high content of titanium and showed signs of alteration in the form of tiny granulations within the cracks dissecting the grains. Oxidation of the mineral must have been minimal and can be designated as class 1 [8]. This is indicated by the occurrence of sulphide and ulvöspinel grains which would otherwise have been oxidized to ilmenite.

Microscopic studies of polished sections of the material from site *Y* also revealed the presence of homogeneous titanomagnetite. The magnetic grains were, however, much smaller than those of sites *W* and *X*, their diameters averaging about 5 μ. Locally some tiny needles of hematite as well as of ilmenite were seen, but skeletal crystals of homogeneous titanomagnetite typical for lavas were predominant.

6. Paleomagnetic Data

The results of measurements of the virgin directions of magnetization are presented in Fig. 1A in the form of stereograms plotted on an equal area net. Range and median of the virgin intensity of remanent magnetization are listed in Tab. 2.

Table 2. *Intensities of NRM in* 10^{-3} *emu/cm*^{3}

Site	Range	Median
W	0.4— 7.2	0.8
X	0.7— 1.2	1.0
Y	3.3—11.7	4.5

One of the two above mentioned laboratory "cleaning" techniques was applied to each specimen. Altogether 15 specimens, that means 5 specimens from each site, were "cleaned" by the AF demagnetization procedure. The maximum field intensity was stepwise increased from 22 to 44, 88, 175 and 350 oe. Specimens that continued to

have an intensity of magnetization greater than 10 % of the original NRM were treated also with peak fields of 700 and 1400 oe.

The remaining 28 specimens were subjected to thermal demagnization and were heated successively to 200°, 300°, 400°, 500°, 520° and 560° C. For the two lower heating steps the heating was con-

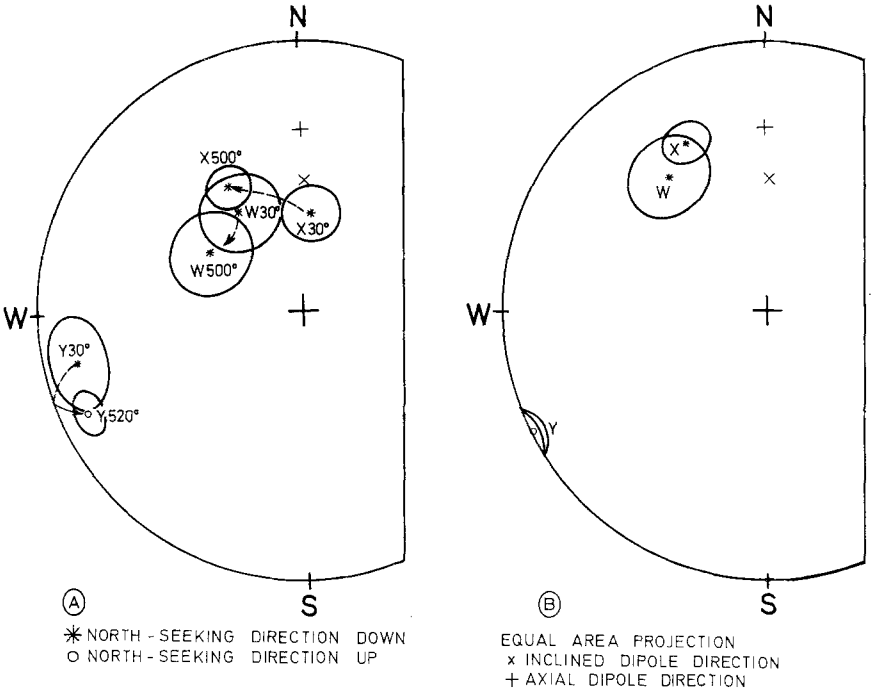


Fig. 1. Group mean directions of magnetization and their 95 percent confidence circles: A) Change in direction due to thermal demagnetization procedure; B) Group mean directions after applying tectonic corrections

ducted with only 8 pilot specimens. The heating up to 560° C was also performed with only these 8 specimens because the stable direction was already achieved by heating the samples to 500° C (sites *W* and *X*) and 520° C (site *Y*) respectively. Application of the "cleaning" procedures resulted in the change of the directions of magnetization in the specimens from all three sites. This shows that the original magnetization was masked by secondary components probably of viscous origin. In the case of sites *W* and *X* the thermal "cleaning" procedure was more effective than treatment with alternating fields. This is a behavior unusual for most igneous rocks, though the same effect has previously been observed in some

Jamaican Upper Cretaceous intrusions and lavas [9]. The phenomenon can be interpreted in terms of the large sizes of the homogeneous titanomagnetite grains which are responsible for the low coercivity of the rock and cause instability of the NRM [10]. As a result of the low coercivity the primary thermoremanent magnetization is also affected by the AF demagnetizing process.

Specimens of site *Y* were of course magnetically much more stable, because their magnetic constituents occurred in the form of very small grains of homogeneous titanomagnetite. In this case demagnetization by thermal and AF treatment produced equally good results.

The mean directions of magnetization for sites *W* and *X* were determined mainly from the results of the thermal "cleaning" process, but some selected data for specimens treated by the AF procedure were also included. In the case of site *Y* all AF demagnetized specimens could be used for the determination of the mean direction of magnetization. A few specimens from sites *W* and *X* revealed a very low magnetic stability and an erratic behavior on demagnetization and were, therefore, excluded from final analysis. Criteria for exclusion were the scatter of directions and the behavior during partial demagnetization [11]. For the AF demagnetized specimens which were accepted, the optimum peak field value was chosen on the basis of minimum within-site dispersion [12].

The directions after AF and thermal "cleaning" are listed in Tab. 3, section b, and are also shown in Fig. 1.

Table 3. *Site Mean Directions and Statistical Parameters*

Site	*N*	*D*	*I*	*R*	*k*	α_{95}
a) Mean virgin directions						
W	15	327.9°	54.0°	13.752	11.2	12.0°
X	16	5.9°	60.3°	15.168	18.0	8.9°
Y	13	258.3°	14.4°	12.164	14.3	11.3°
b) Mean directions after demagnetization						
W	13	302.8°	56.6°	11.998	12.0	12.5°
X	12	329.9°	45.5°	11.753	44.6	6.6°
Y	13	245.9°	−12.0°	12.783	55.2	5.6°

The directions before applying tectonic corrections are shown on the left side of Fig. 1. The mean virgin directions are those corresponding to the temperature of 30° C. The directions obtained by thermal demagnetization at 500° and 520° are also shown. For each mean direction the 95 % circle of confidence is given. The data from

all three sites show a change in the direction of magnetization; the least amount of change occurs for site W and the greatest change for site X. The changes are away from the direction of the present day geomagnetic field. Site Y changes from positive to negative inclination. The effect of the low coercivity of the magnetic constituents is conspicuous in the specimens of site W, were the 95 % radius of confidence has increased slightly after thermal "cleaning".

On the right of the Figure the "clean" mean directions are shown after applying the tectonic corrrections given in Tab. 1.

7. Paleomagnetic Pole Positions

Fig. 2 and Tab. 4 summarize the results in terms of paleomagnetic pole positions.

Table 4. *Average Paleomagnetic Pole Positions*

Site	Paleomagnetic Pole		Ellipse of Confidence	
	λ'	Φ'	δp	δm
W	154.3° W	55.5° N	9.4°	15.0°
X	164.6° W	65.3° N	4.4°	7.4°
Y	158.4° W	25.7° S	2.8°	5.6°

The paleopole positions for all three sites differ from each other and that for Y differs significantly from the other two. Unfortunately the precision parameters for W and X are significantly different and hence it is not possible to test whether the two directions also differ significantly [13]. The paleopole for W does not, however, lie within the confidence oval for X and it is, therefore, best to discuss all three paleopoles separately.

The paleopole from the fresh rock at site W is close to the mean pole position *UC* previously found for the Upper Cretaceous in Jamaica [9, 14]. The latter has been derived from the data of eleven sites of probably Maestrichtian age ($70 \cdot 10^6$ y). The two pole positions are similar and it seems, therefore, that site W reflects the direction of the Lower Cretaceous geomagnetic field for Jamaica. This pole position differs significantly from the position of the mean Cretaceous pole for North America NA (the latter, derived from eight different localities, has $\alpha_{95} = 2.4^0$).

The pole for site X is intermediate between the Upper Cretaceous pole *UC* for Jamaica and the present day pole. This can be interpreted in several ways. One explanation is that the "cleaning" of

the specimens from site *X* was incomplete as a result of the low stability of the magnetic minerals, discussed in section 5. But the data from this site show the greatest change during demagnetization and a satisfactory decrease of the radius of the circle of confidence as shown in Fig. 1. Likewise it is possible that the tectonic correction used for this site was not quite corret. Further, it is possible that the difference in pole positions between sites *W* and *X* can be

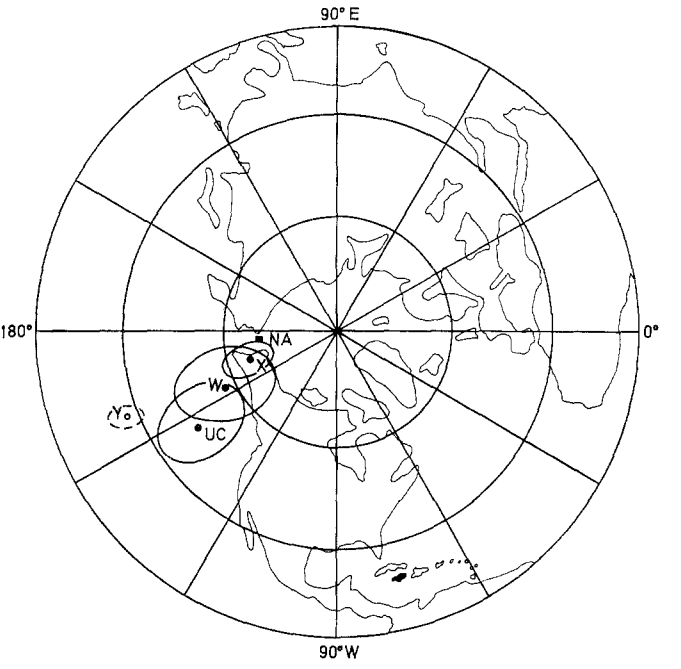


Fig. 2. Paleomagnetic pole positions calculated from mean directions of magnetization and corresponding circles of confidence. *NA* denotes the Cretaceous pole position for North American and *UC* the position of the Upper Cretaceous paleopole for the Caribbean according to [9, 14]

Solid circles: Upper hemisphere; open circles: Lower hemisphere

interpreted as the normal scatter in the directions of different sites. The data available are, however, for two sites only and, therefore, the statistical tests cannot decide whether this hypothesis is acceptable.

The low latitude position of the paleomagnetic pole for site *Y* can be interpreted as the result of a non axial dipole configuration during a polarity change of the geomagnetic field. Polarity changes by a flip over of the dipole axis seem to have occured several

times in the earth's history [15, 16] and have also been observed in Upper Cretaceous rocks on Jamaica [9]. Low latitude or equatorial pole positions can be taken as the instantaneous records of the non-axial dipole configuration during such a type of field reversal. Paleomagnetic data for Lower Cretaceous [17] indicate that the polarity change occurred $110 \cdot 10^6$ y ago at the close of Aptian and beginning of Albian. This age is in agreement with the age of the lavas deduced from the fauna of associated sediments. The proposed polarity change supports thus the age assigned to the lavas on the basis of paleontological data. According to Tab. 2 the NRM at site Y has a comparatively high intensity. This seems to indicate that during the polarity change the moment of the non-axial dipole was comparable in magnitude with the moment of the normal geomagnetic dipole parallel to the earth's axis of rotation.

An alternative explanation for the Lower Cretaceous pole distribution, given in Fig. 2, would be the assumption of large scale rotations of Jamaica. This hypothesis is not favored in view of the general east-west structural trends in the northern Caribbean [6]. Since paleomagnetic results from Puerto Rico [18] also yield a pole position close to *UC*, the rotation hypothesis would require simultaneous large scale rotations of both islands, a tectonic process which could not easily be reconciled with the structural configuration of northern Caribbean.

We hold this view even though MacDonald and Opdyke [19] have recently suggested a large scale rotation of Jamaica since the Cretaceous. They relate their paleomagnetic data for the Cretaceous rocks of the Guajira peninsula in northern Colombia to those of Jamaica [9] and propose a large scale rotation of Jamaica, Puerto Rico and Guajira. The Guajira peninsula lies, however, outside the main Caribbean area and would be expected to have had a tectonic history different from that of the northern Caribbean. We do not, therefore, believe that it is legitimate to interpret paleomagnetic data from Jamaica in terms of the tectonics of northern South America.

8. Discussion

The similarity between the Upper Cretaceous paleopole obtained from eleven rock units and the Lower Cretaceous pole for site W allows the determination of the "normal" Jamaican Cretaceous paleopole for 12 rock units. The resulting mean paleopole is situated at 146.6^0 W, 44.9^0 N and has an ellipse of confidence with semi-axes 8.7^0 and 13.0^0, and the precision parameter $k = 19.1$. The pre-

cision parameter as well as the ellipse of confidence show some minor improvements in comparison with the result based on Upper Cretaceous rocks alone [14].

In terms of plate tectonics this pole position should also be valid for the whole Caribbean plate. It can, therefore, be accepted as a verification of Fink and Harrison's [18] Upper Cretaceous pole (age of the relevant rock units about $95 \cdot 10^6$ y) for Puerto Rico situated at 149.2° W, 35.1° N. Although the result is based on data from a single Lower Cretaceous rock unit, it can be regarded as evidence that the Caribbean has not moved during the Cretaceous age over distances which are detectable by the paleomagnetic technique. Moreover, it suggests also that there was no appreciable polar wander between the Aptian and Maestrichtian periods. The Lower Cretaceous result supports the tectonic model put forward by Guja and Vincenz [9] and relating to the movement of the Caribbean plate since Upper Cretaceous. According to this concept the Caribbean plate has moved from the south-west relative to its present day position through 1000 km. The displacement of Jamaica can be described in present day coordinates as having a northward component of 600 km and an eastward component of 700 km. Using geochronological data for some of the Upper Cretaceous rock units studied [20], it follows that this displacement occurred during the last $70 \cdot 10^6$ y and corresponds to the minimum average velocity of 1.4 cm/y [9]. This result is consistent with other geophysical data from the Caribbean. Earthquake focal mechanism studies confirm the plate behavior of the Caribbean region and also its present-day easterly movement [21]. Magnetometer surveys in southern Caribbean have revealed magnetic anomaly patterns with typical oceanic lineation in the east-west direction [22, 23], suggesting that sea-floor spreading occurred also in the north-south direction. There is strong evidence that the Caribbean plate was originally a part of the East Pacific geotectonic plate system [24]. Thus to provide evidence for plate movement in the proper direction the study of the Pacific spreading rift zones should be intensified. The necessary driving source might be identified as the east-west oriented Galapagos rift zone [25] and the roughly north-south trending East Pacific Rise.

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